PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

Title of your Project

EV Charging Station Locator & Booking

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Project Guide

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1. Introduction

Introduction

The project is a **Smart EV Charging Locator and Booking System**.

It helps electric vehicle users **find nearby charging stations and check availability easily**.

Problem Identified

1 EV users face difficulty **locating nearby charging stations**.

2 Lack of **real-time information about slot availability** causes waiting and inconvenience.

Solution Proposed

1 The system **locates nearby EV charging stations using GPS**.

2 It **shows real-time availability of charging slots** to help users plan charging.

2. Objectives

1.To help electric vehicle users easily locate nearby charging stations using location-based services.

2.To provide real-time information on charging slot availability at different stations.

3.To enable advance booking of charging slots to reduce waiting time for users.

4.To improve the efficient utilization of charging station infrastructure through optimized scheduling.

3. Scope

1. **Can be applied in cities and highways** where electric vehicle charging stations are available.

2. **Can be useful to electric vehicle owners** to easily locate nearby charging stations.

3. **Can help charging station operators** manage charging slots and reduce congestion.

4. **Can be useful in smart city and sustainable transportation systems** to support the growth of electric vehicles.

4. Feature /Functionality

Feature 1: Locate Nearby Charging Stations

Uses GPS to show nearby EV charging stations and their details.

Feature 2: Real-time Slot Availability

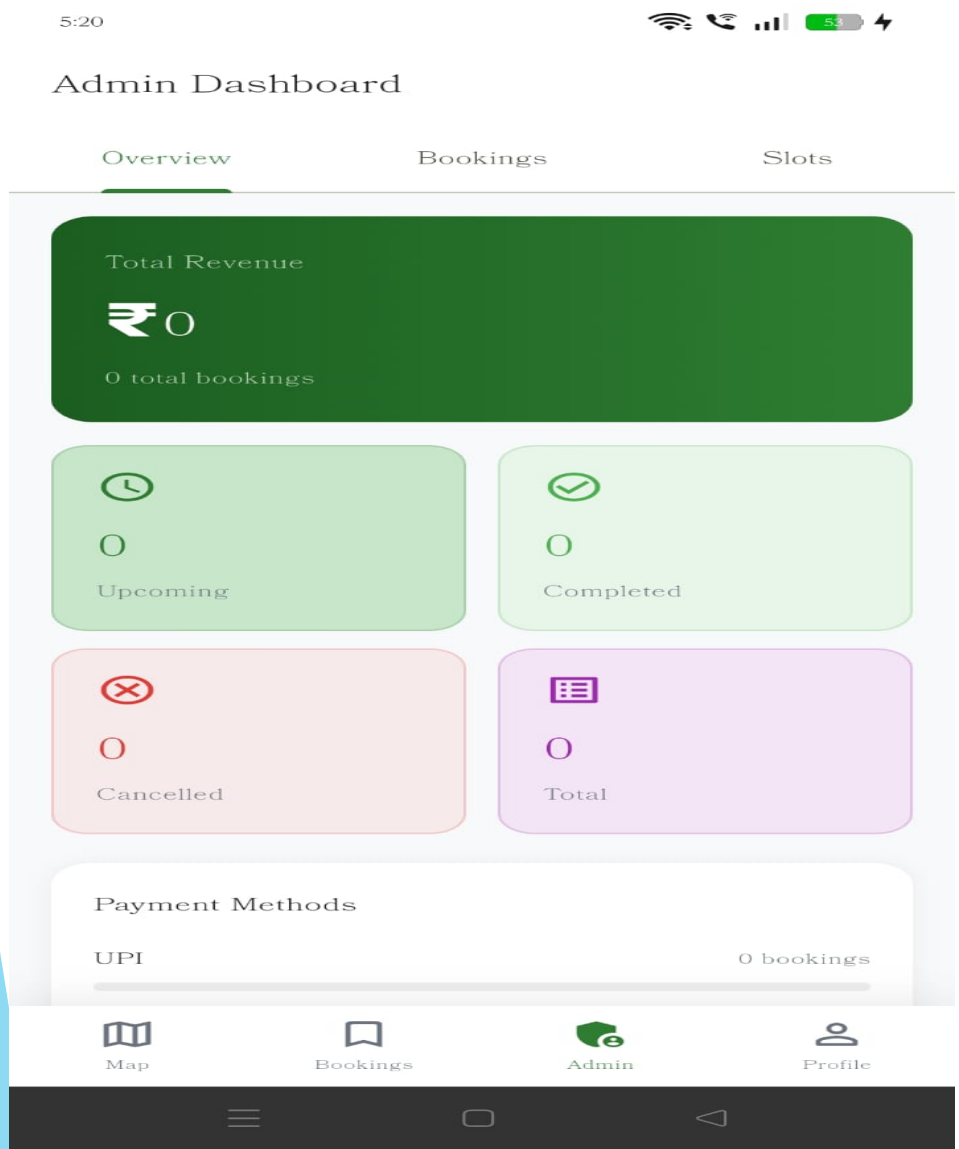
Displays real-time information about available and occupied charging slots.

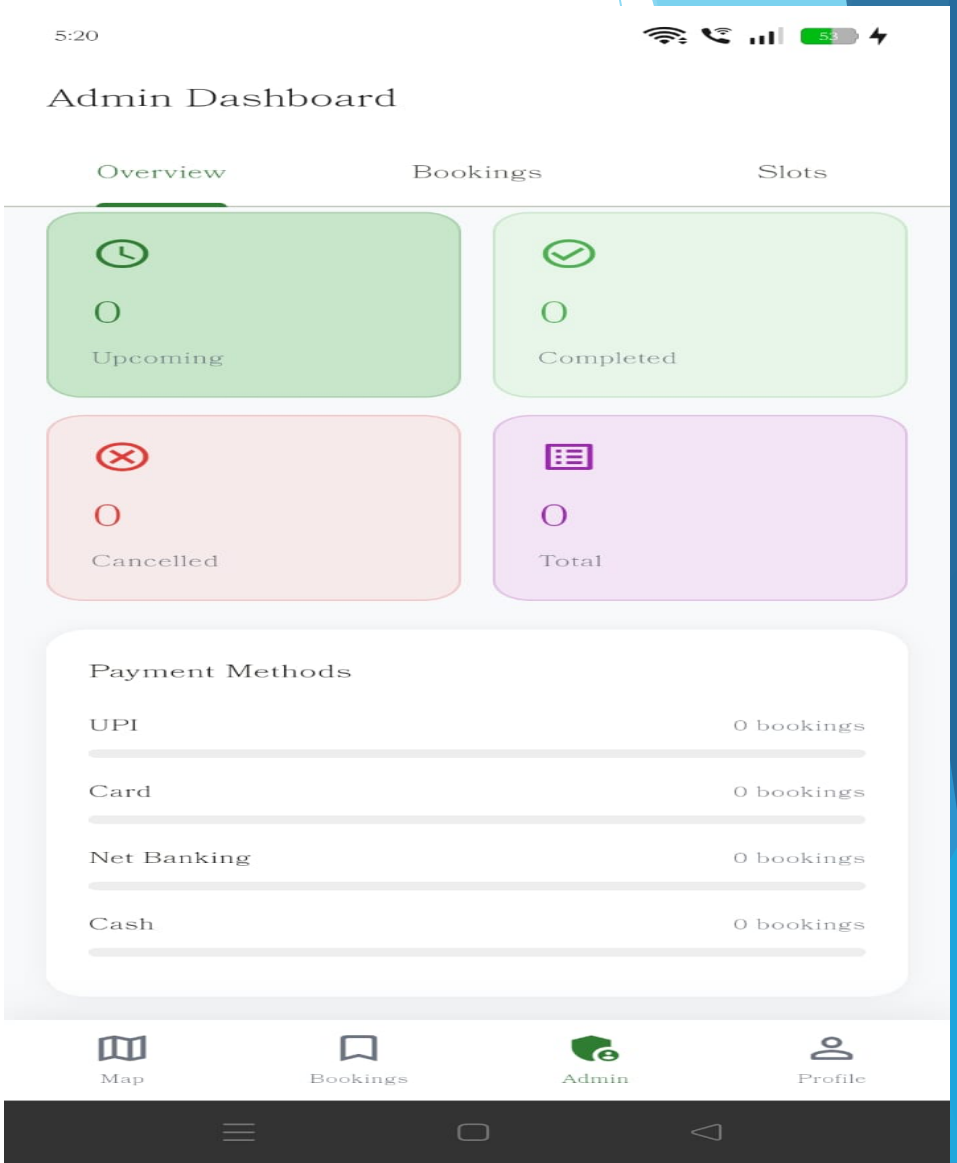
Feature 3: Advance Slot Booking

Allows users to book charging slots in advance to reduce waiting time.

5. Outcome of Project

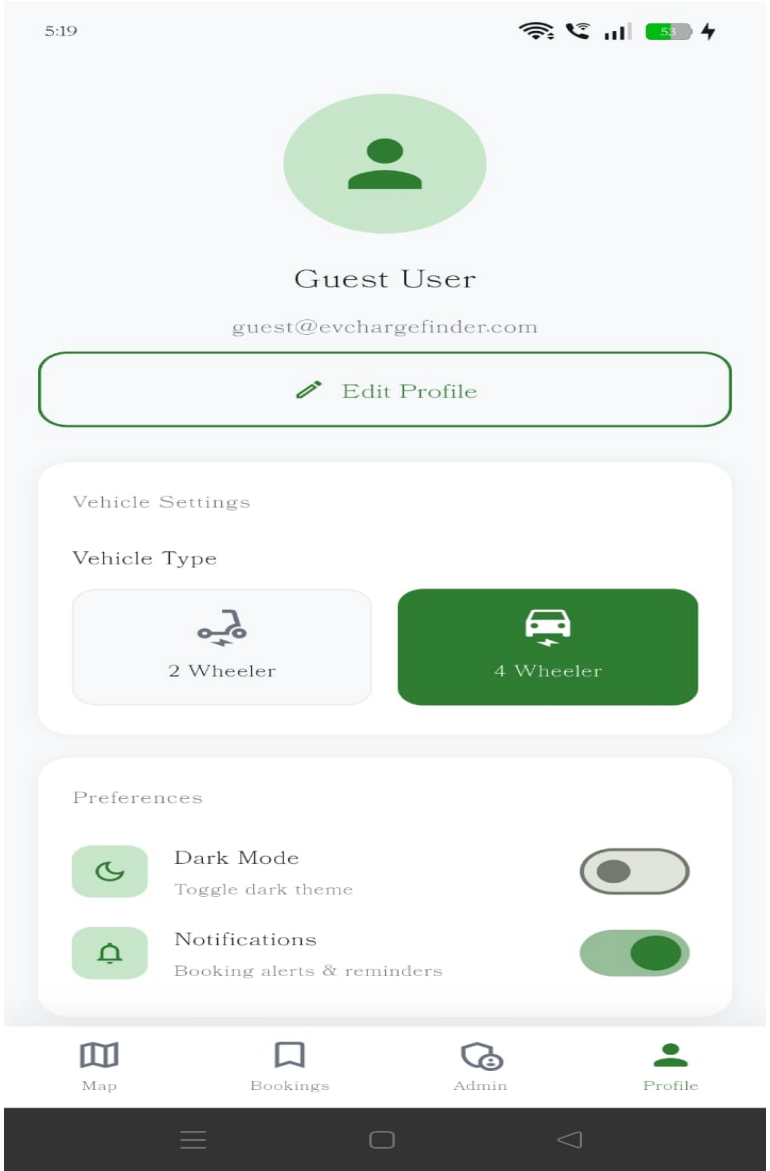
1. Admin dashboard

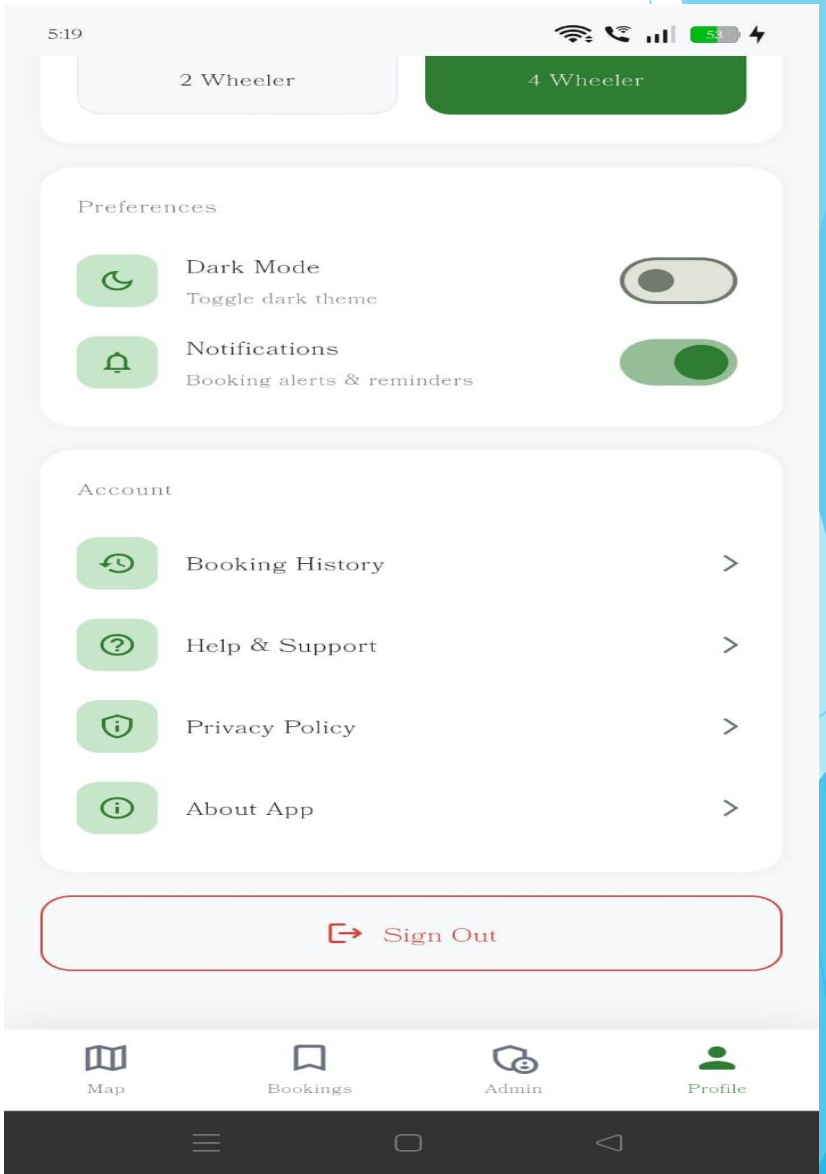




5. Outcome of Project

2. User profile





6. Technology Stack

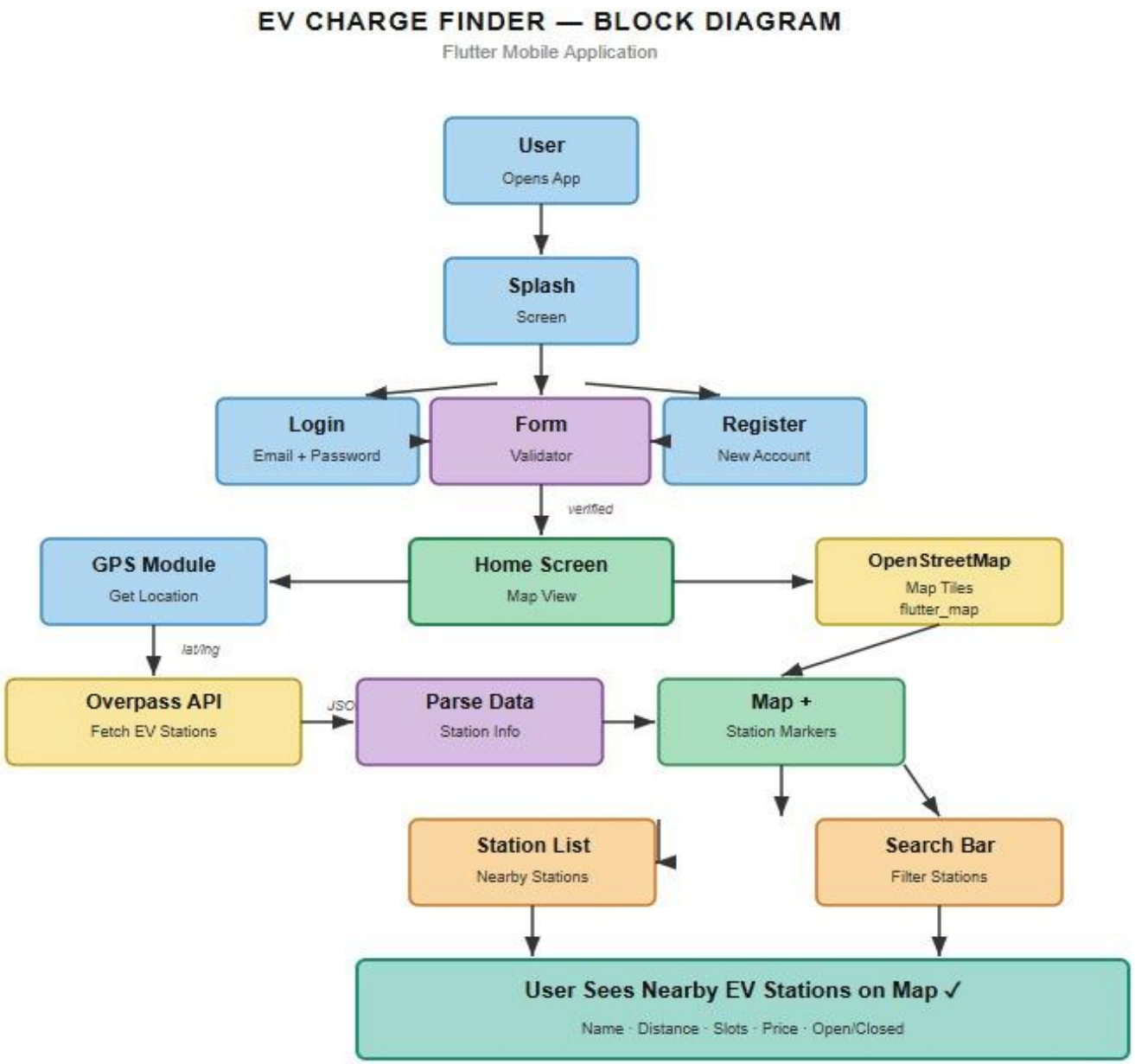
Frontend (GUI) Development

1. **Flutter Framework**: Used for developing a high-performance, cross-platform mobile application.
2. **Dart Language**: Utilized for application logic and seamless UI rendering.
3. **Google Maps API**: Integrated to provide interactive map views and location-based station searching.

Backend & Database

1. **Firebase Authentication**: Handles secure user registration and login processes.
2. **Firebase Firestore**: Serves as the real-time NoSQL database for station data and slot availability.
3. **Location Services**: Utilizes GPS to fetch and display stations based on the user's live coordinates.

7. Block Diagram



SDGs as per the UN Framework

Primary SDGs

- **SDG 7: Affordable & Clean Energy**
  - Promotes EV usage and clean energy
  - Reduces dependence on fossil fuels

- **SDG 9: Industry, Innovation & Infrastructure**
  - Develops smart EV charging platform
  - Supports modern infrastructure

SDGs as per the UN Framework

Secondary SDGs

- **SDG 11: Sustainable Cities**
  - Reduces pollution and supports green cities
- **SDG 13: Climate Action**
  - Helps lower carbon emissions
- **SDG 8: Economic Growth**
  - Creates opportunities for charging services
- **SDG 12: Responsible Consumption**
  - Optimizes energy usage with real-time data

Thank You...!!